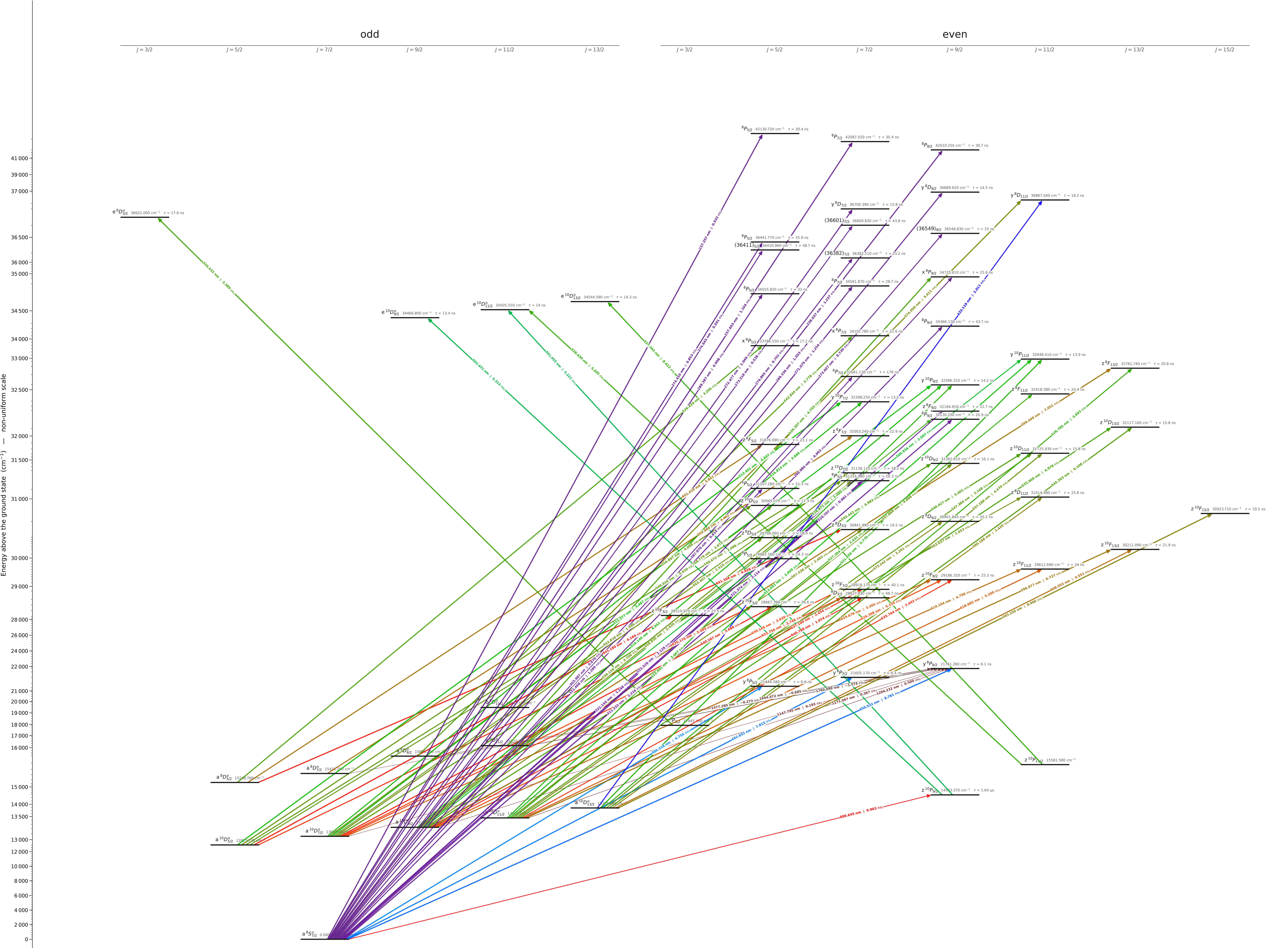
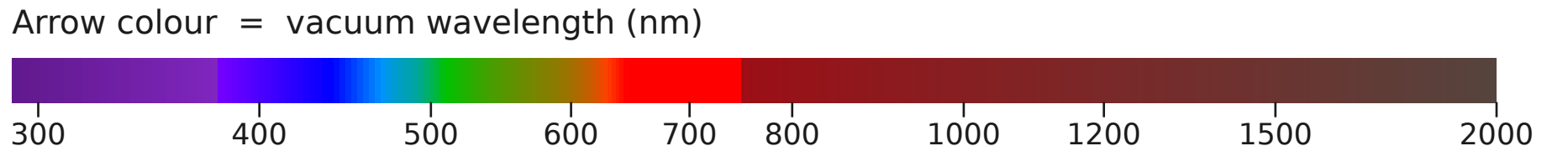


Europium

Eu (Eu I). Strongest classified transitions below 2 μm (NIST): 68 levels, 90 lines.

Eu + ionisation limit 45734.74 cm⁻¹ = 5.67039 eV (218.652 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element |⟨J||er||J'⟩| in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST), or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
a⁴S _{3/2} - y⁶P _{9/2}	459.5322	459.4035	652.3862	8.781	1.610e+08	26.09	99.9	broad blue line
a⁴S _{5/2} - y⁶P _{7/2}	462.8522	462.7226	647.7067	7.815	1.560e+08	24.87	99.8	strong blue resonance line of the y8P triplet
a⁴S _{5/2} - y⁶P _{5/2}	466.3183	466.1878	642.8923	6.756	1.520e+08	24.11	100	strong blue resonance line of the y8P triplet
a¹⁰D _{9/2} - y¹⁰P _{11/2}	506.9356	506.7943	591.3817	2.097	5.700e+06	11.45	7.92	507 nm: optical pumping a10D9/2
a¹⁰D _{7/2} - y¹⁰P _{11/2}	513.0520	512.9091	584.3315	3.959	1.960e+07	11.45	27.2	513 nm: optical pumping a10D11/2
a¹⁰D _{5/2} - y¹⁰P _{11/2}	521.6558	521.5105	574.6940	5.770	3.960e+07	11.45	55	521.5 nm: decay channel y10P11/2 -> a10D13/2 that populat...
a¹⁰D _{3/2} - z¹⁰F _{13/2}	583.2594	583.0977	513.9951	8.966	5.130e+07	8.162	100	quasi-cyclic 583 nm transition of the metastable a10D13/2...
a¹⁰D _{11/2} - z¹⁰F _{13/2}	596.8767	596.7114	502.2687	6.727	3.080e+07	7.267	67.5	decay channel z10F13/2 -> a10D11/2 used in the pump-back
a¹⁰D _{5/2} - z¹⁰F _{13/2}	608.5535	608.3850	492.6313	4.551	1.330e+07	7.267	29.1	609 nm
a⁴S _{5/2} - z¹⁰P _{9/2}	686.6448	686.4554	436.6048	0.963	5.800e+05	0.09705	97.3	narrow-line "red" MOT at 687 nm
a¹⁰D _{7/2} - y⁶P _{9/2}	1147.7946	1147.4805	261.1900	0.193	5.000e+03	26.09	0.0032	optical leak of the 460 nm line
a¹⁰D _{9/2} - y⁶P _{9/2}	1171.0671	1170.7467	255.9994	0.367	1.700e+04	26.09	0.0108	optical leak of the 460 nm line
a¹⁰D _{11/2} - y⁶P _{9/2}	1204.2317	1203.9022	248.9492	0.500	2.900e+04	26.09	0.0178	optical leak of the 460 nm line
a⁴D _{5/2} - y⁶P _{9/2}	1577.2846	1576.8537	190.0687	0.273	3.860e+03	26.09	0.0024	optical leak of the 460 nm line
a⁴D _{3/2} - y⁶P _{9/2}	1644.4718	1644.0227	182.3032	0.685	2.140e+04	26.09	0.0133	optical leak of the 460 nm line
a⁴D _{11/2} - y⁶P _{9/2}	1760.0986	1759.6180	170.3271	1.574	9.210e+04	26.09	0.0572	optical leak of the 460 nm line

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πν).

¹⁵³Eu hyperfine structure (I = 5/2)			
Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
a⁶S _{7/2}	-8.85320(20)	-1.785(4)	F=1: +98.64 F=2: +81.45 F=3: +55.46 F=4: +20.45 F=5: -23.87 F=6: -77.91

Measured A, B constants; sources in the data file.